

Formation Control of Nonlinear Planar Mobile Robots

Maurizio Melillo

DAUIN

Politecnico di Torino

s360164@studenti.polito.it

Lorenzo Bucaletti

DAUIN

Politecnico di Torino

s353632@studenti.polito.it

Andrea Sangineto

DAUIN

Politecnico di Torino

s362052@studenti.polito.it

Michele Savino

DAUIN

Politecnico di Torino

s361490@studenti.polito.it

Abstract—This paper presents the design and simulation of a distributed cooperative control architecture for a heterogeneous multi-agent system of three planar mobile robots required to maintain a rigid equilateral-triangular formation while tracking a moving reference trajectory. Because the agents are governed by distinct nonlinear affine dynamics, a two-stage hierarchical strategy is adopted: Input-Output Feedback Linearization (IOFL) first cancels each agent's nonlinearities and maps it onto a uniform linear double-integrator virtual plant; distributed formation protocols then operate on this homogeneous network. Two cooperative methodologies are investigated and compared. The *Loop-Shaping* approach designs a dynamic output-feedback compensator $K(s)$ in the frequency domain, relying solely on relative position measurements and enforcing stability via the Nyquist criterion on the pinned modal loop functions. The *State-Variable Feedback* (SVFB) approach derives an optimal LQR gain through an Algebraic Riccati Equation and couples it with a Luenberger–Kalman observer to reconstruct full state information from noisy position measurements. Both methods are evaluated through systematic parameter sweeps covering LQR weighting matrices, coupling gains, observer tuning, input saturation, output disturbances, network topologies, and reference trajectories (linear, circular, sinusoidal). Experimental results demonstrate that SVFB achieves faster convergence and near-zero steady-state errors under ideal conditions, while the Loop-Shaping approach provides inherently superior robustness against persistent disturbances without requiring state estimation.

Index Terms—formation control, multi-agent systems, input-output feedback linearization, linear quadratic regulator, Luenberger observer, loop shaping, distributed consensus, pinning control, mobile robots

I. INTRODUCTION

The main objective of this project is to design, simulate, and analyze a distributed cooperative control architecture for a multi-agent system composed of three heterogeneous planar mobile robots. The network of agents is required to autonomously cooperate in order to achieve and maintain a specific rigid geometric shape, that is, an equilateral triangular formation, while simultaneously moving along a predefined target trajectory.

This problem presents significant structural and control challenges. The three agents are heterogeneous and their dynamic behaviors are governed by non-linear affine equations, featuring complex phenomena. Directly applying linear distributed consensus protocols over such diverse and non-linear models is practically infeasible.

To overcome this, the project adopts a rigorous two-stage hierarchical control strategy:

- **Local Input-Output Feedback Linearization (IOFL):** In the first stage, the non-linearities of each individual robot are mathematically canceled out at the local level. By deriving the exact drift terms and decoupling matrices, an IOFL control law is applied to map each unique physical robot into an identical, strictly linear double-integrator model. This operation completely removes the zero dynamics and homogenizes the fleet, paving the way for linear multi-agent cooperative protocols.
- **Distributed Formation Control:** Once the homogeneous linear virtual plants are obtained, the agents are modeled in order to coordinate using only local information shared across a communication graph. To ensure that the formation behaves as a single rigid body tracking a moving reference, a tracking anchor (pinning) is applied to the network.

To provide a comprehensive analysis of the formation control problem, this report investigates and compares two distinct cooperative methodologies:

- 1) **State Variables Feedback (SVFB) Approach:** A time-domain cooperative protocol based on the optimal Linear Quadratic control theory. By using both relative position and velocity tracking errors, an Algebraic Riccati Equation (ARE) framework is developed, and thanks to Lyapunov stability conditions, is computed a static feedback gain matrix that synchronizes the overall network.
- 2) **Loop-Shaping Approach:** A dynamic output-feedback protocol formulated in the frequency domain. This approach relies exclusively on the relative output errors (measured positions) among neighbours. The network's spatial disagreement is stabilized by designing a dynamic compensator $K(s)$ that regulate the behaviour of the formation of agents in order to simultaneously follow the desired trajectory and reject some external disturbances.

The remainder of this report is structured as follows. Section II details the mathematical derivation of the exact feedback linearization for the three heterogeneous agents. Sections III and IV present the theoretical design, Simulink/MATLAB implementation, and experimental graphical analysis of the

Loop-Shaping and State-Feedback controllers, respectively. Finally, Section V offers a comparative analysis of the two approaches, discussing their robustness, disturbance rejection capabilities, and overall performance.

II. INPUT-OUTPUT FEEDBACK LINEARIZATION

The first fundamental stage of the proposed control architecture is to cancel the heterogeneous nonlinearities characterizing the three mobile robots. Each agent is initially modeled with non-linear affine dynamics, where the measured output corresponds to the planar position.

To apply the Input-Output Feedback Linearization (IOFL), the output of each agent is differentiated twice with respect to time until the physical control input (the acceleration commands) explicitly appears in the equations. This mathematical process reveals that the vector relative degree for each agent is exactly two for both spatial coordinates. The sum of these relative degrees is four, which perfectly matches the total dimension of the state vector. This structural property is of paramount importance: it ensures that the feedback linearization covers the entire system dynamics, effectively meaning there are no hidden or zero dynamics to stabilize.

Once the drift vector and the decoupling matrix are extracted, and assuming the latter is non-singular, the exact feedback linearizing control law can be applied. This operation successfully transforms each distinct physical robot into a standardized, identical linear double-integrator model, which can then be driven by a newly defined virtual input.

A. Mathematical Derivation for the Agents

To successfully implement the IOFL, the specific drift terms and decoupling matrices were analytically derived for all three agents based on their physical models. A critical step was to rigorously verify the invertibility of each decoupling matrix over the entire state space to guarantee that the control law could always be computed safely.

The main findings of this derivation are summarized as follows:

- **Agent 1:** The physical equations naturally separate without cross-axis interactions. The resulting decoupling matrix is completely constant and diagonal. Its determinant is trivially non-zero, ensuring immediate global invertibility.
- **Agent 2:** The dynamics yield a diagonal decoupling matrix whose elements depend on the planar position. However, because these variable elements are governed by squared trigonometric functions (sine and cosine), the denominators are bounded and never reach zero. This analytical property proves that the matrix determinant remains strictly positive everywhere.
- **Agent 3:** This agent presents the highest complexity due to cross-axis interactions, resulting in a full, non-diagonal decoupling matrix. By analytically computing its determinant, we demonstrated that the product of the

strictly positive main diagonal elements always mathematically dominates the bounded off-diagonal cross-products. This guarantees that the matrix remains universally non-singular despite the complex non-linear couplings.

By successfully isolating and proving the invertibility of these matrices, the exact local control actions were reliably formulated to cancel the physical nonlinearities, allowing the distributed formation algorithms to operate on simple linear virtual plants.

III. DISTRIBUTED CONTROL: STATE-FEEDBACK APPROACH

The first cooperative control strategy investigated is based on state-feedback approach. This section presents the design and implementation of the distributed state-feedback controller adopted for the multi-agent formation problem. The proposed control strategy combines optimal state-feedback with state estimation to achieve accurate trajectory tracking while preserving the desired formation. The controller design procedure, implementation details, and experimental validation are described in the following subsections.

A. Controller Design

In the State Variables Feedback (SVFB) approach, was designed a distributed control protocol based on the Linear Quadratic Regulator (LQR) optimal control theory in order to stabilize this linear virtual plant and achieve the desired formation tracking.

The optimal feedback gain matrix K is computed by solving the Algebraic Riccati Equation (ARE) using carefully selected weighting matrices Q and R . The matrix $Q \in \mathbb{R}^{4 \times 4}$ penalizes the state tracking errors (both planar positions and velocities), while $R \in \mathbb{R}^{2 \times 2}$ penalizes the virtual control effort.

Since only the planar positions (x and y) are measured (defined by the output matrix C), the full state vector is not directly available for feedback. To overcome this, a Luenberger Observer was designed to estimate the velocities. The observer gain F is similarly computed by solving a Riccati equation for the dual system using the tuning matrices Q_o and R_o , allowing for an optimal balance between convergence speed and sensor noise rejection.

B. Implementation and Workflow

The discrete-time implementation of the SVFB architecture was developed in MATLAB/Simulink using a Forward Euler integration scheme with a time step of $dt = 0.01$ seconds over a 20-second simulation window.

The execution workflow at each time step follows a strict sequence:

- 1) **Measurement & Estimation:** Position measurements are acquired, potentially corrupted by additive Gaussian noise and sinusoidal disturbances. The Luenberger observer dynamically updates the state estimate $\hat{x}_i$ for each agent.

- 2) **Distributed Protocol Evaluation:** Using the estimated states, the local formation error ε_i is computed based on the adjacency matrix Adj and the target geometric offsets H . The virtual control input ν_i is generated as a sum of the cooperative formation term ($c \cdot K \cdot \varepsilon_i$) and, exclusively for the leader, the tracking term ($c_l \cdot K \cdot (x_r - \hat{x}_1)$) plus a feedforward term.
- 3) **Inverse Feedback Linearization:** The virtual input ν_i is mapped back into the physical control input u_i via the local `fb_lin_agent` inverse functions, which compute the required non-linear cancellation using the estimated states.
- 4) **Saturation & Integration:** Both virtual and physical control inputs can be strictly saturated to NU_MAX and U_MAX respectively, simulating real-world actuator limits, before updating the physical plant dynamics.

C. Method-Specific Experiments and Analysis

To properly tune the system and analyze its behavior under varying conditions, an extensive experimental campaign was conducted. The tuning process systematically explored the variation of the LQR weights (Q, R), the coupling gains (c, c_l), the observer parameters (Q_o, R_o), and the impacts of topological, physical, and environmental constraints.

Baseline Setup

Experiment 1:

- **Topology:** Undirected ($Adj = [0, 1, 1; 1, 0, 1; 1, 1, 0]$)
- **LQR Weights:** $Q = \text{diag}([10, 1, 10, 1]); R = \text{eye}(2)$
- **Coupling Gains:** $c = 1; c_l = 3$
- **Saturation/Noise/Disturbance:** OFF
- **Trajectory:** Linear

Analysis: The baseline test establishes the nominal behavior of the system. It achieved a tracking convergence time of 2.49 s and a formation convergence time of 4.33 s, with zero steady-state tracking error.

R & Q Matrix Tuning

Experiment 2 - Q tuning :

- **Topology:** Undirected
- **LQR Weights:** $Q = \text{diag}([50, 1, 50, 1]); R = \text{eye}(2)$
- **Coupling Gains:** $c = 1; c_l = 3$
- **Saturation/Noise/Disturbance:** OFF

Analysis: Increasing the Q matrix prioritizes tracking precision over energy savings. This adjustment significantly accelerated the tracking convergence to 1.61 s. However, this rapidity requires a critical actuation effort, nearly doubling the peak input for Agent 1 (up to 75.20 N).

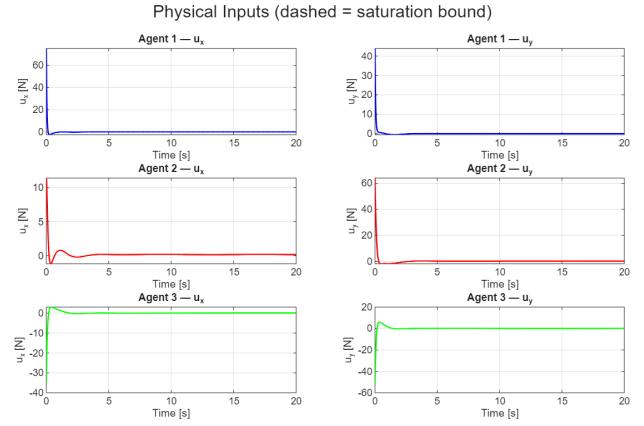


Fig. 1: Results for Experiment 2 demonstrating increased control activity.

Experiment 3 - R tuning:

- **Topology:** Undirected
- **LQR Weights:** $Q = \text{diag}([10, 1, 10, 1]); R = 5 \cdot \text{eye}(2)$
- **Coupling Gains:** $c = 1; c_l = 3$
- **Saturation/Noise/Disturbance:** OFF

Analysis: Increasing the R penalty successfully reduces the required control input activity. This conservative approach, however, penalizes the system's responsiveness, noticeably increasing the tracking convergence time to 3.65 s and the formation convergence time to 6.46 s.

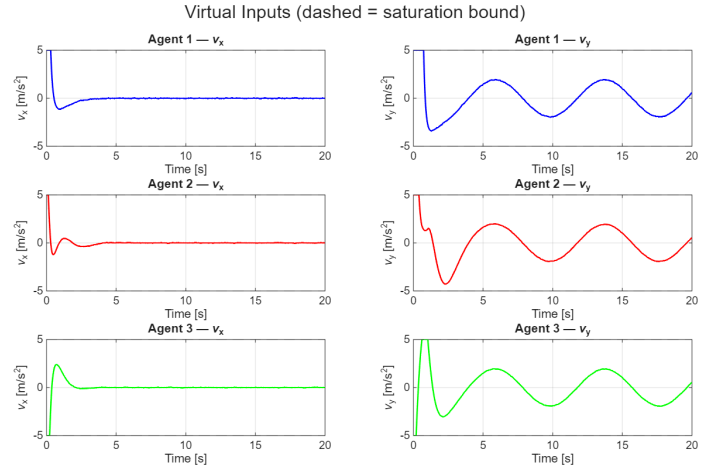


Fig. 2: Results for Experiment 3 demonstrating reduced control effort.

Leader Coupling Gain - tuning analysis(c_l) & (c)

Experiment 4:

- **Topology:** Undirected
- **LQR Weights:** Baseline (Q_{base}, R_{base})
- **Coupling Gains:** $c = 1; c_l = 8$
- **Saturation/Noise/Disturbance:** OFF

Analysis: While increasing the leader's tracking gain completely eliminated the steady-state tracking error, it imposed

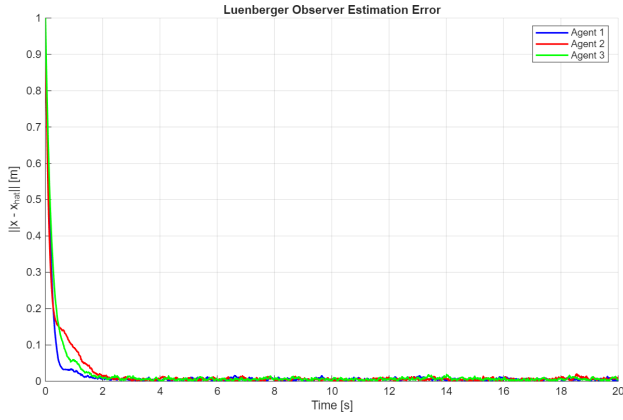


Fig. 3: Results for Experiment 6 with sensor noise.

a substantial control burden on Agent 1. Interestingly, the convergence time did not improve proportionally with the increased effort, indicating an intrinsic limit to the system's dynamic response.

Experiment 5:

- **Topology:** Undirected
- **LQR Weights:** Baseline (Q_{base}, R_{base})
- **Coupling Gains:** $c = 3$; $c_l = 3$
- **Saturation/Noise/Disturbance:** OFF

Analysis: Increasing the inter-agent coupling parameter c reduces both formation convergence time and steady-state error. Nevertheless, this heightened reactivity demands a significantly larger actuation effort from all followers. Henceforth, the baseline $c = 1, c_l = 3$ was maintained as the best trade-off.

Sensor Noise & Observer Analysis

Experiment 6 :

- **LQR Weights:** Baseline (Q_{base}, R_{base})
- **Noise:** ON
- **Observer:** $Q_o = \text{eye}(4) \cdot 0.001$; $R_o = \text{eye}(2) \cdot 0.01$

Analysis: With sensor noise introduced, the observer was initially tuned to favor the internal model reliability over the incoming noisy sensor data. This setup yielded acceptable convergence times (tracking at 2.48 s) but introduced small steady-state errors (formation error: 0.0065 m).

Optimized Observer:

- **LQR Weights:** Baseline (Q_{base}, R_{base})
- **Noise:** ON
- **Observer:** $Q_o = \text{eye}(4) \cdot 0.001$; $R_o = \text{eye}(2) \cdot \text{noise_std}^2$

Analysis: By optimally scaling the observer's R_o matrix strictly to the known noise covariance (noise_std^2), the steady-state tracking and formation errors were more than halved compared to the baseline configuration, despite a slight increase in overall convergence time.

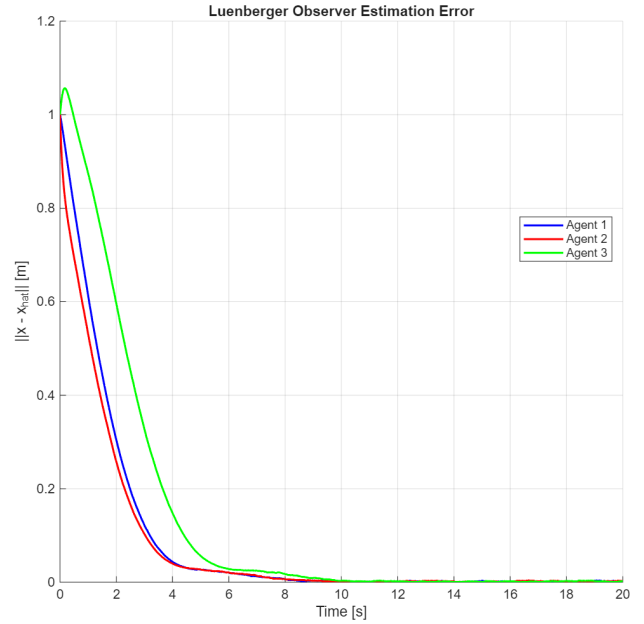


Fig. 4: Results for Experiment 6 showcasing the optimized Luenberger observer.

Impact of Actuator Saturation

Experiment 7:

- **LQR Weights:** Baseline (Q_{base}, R_{base})
- **Saturation:** ON (Virtual $NU_MAX = 5.0$, Physical $U_MAX = 8.0$)
- **Noise:** ON

Analysis: The analysis highlighted a dynamic coupling effect: restricting the virtual input directly limits the physical input, but the relationship is asymmetric. The selected optimal thresholds ($U_MAX = 8.0, NU_MAX = 5.0$) drastically reduced the controller activity without severely penalizing the rapidity of the system's convergence.

Experiment 8 - Output Disturbance:

- **Disturbance:** ON (Amplitude = 0.1 m, Freq = 0.4 rad/s)
- **Noise & Saturation:** ON

Analysis: When subjected to persistent sinusoidal output disturbances, the system experienced severe convergence difficulties. Both steady-state tracking (0.1040 m) and formation errors (0.2340 m) spiked significantly, accompanied by continuous oscillations preventing stabilization within tolerance boundaries.

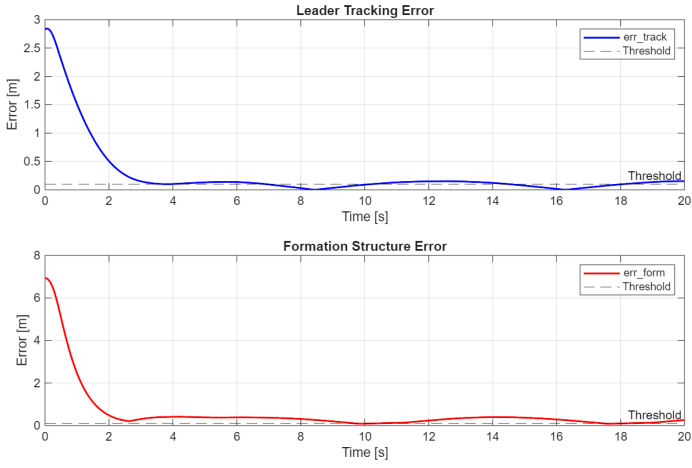


Fig. 5: Results for Experiment 8 under sinusoidal output disturbances.

Topology Variation Analysis

Experiment 9 - $1 \rightarrow 3 \rightarrow 2$:

- **Topology:** Directed cascade ($Adj = [0, 0, 0; 0, 0, 1; 1, 0, 0]$)
- **Saturation & Noise:** ON

Analysis: This directed topology exhibited slightly faster convergence but suffered from a higher steady-state tracking error. This behavior stems from the reduced amount of shared spatial information across the network, demonstrating that fewer corrections are made by the agents, consequently lowering the overall formation precision.

Experiment 10 - $(1 \rightarrow 2 \rightarrow 3 \rightarrow 1)$:

- **Topology:** Directed circle ($Adj = [0, 0, 1; 1, 0, 0; 0, 1, 0]$)
- **Coupling Gains:** $c = 1.16$; $c_l = 3$ (modified for stability)
- **Saturation & Noise:** ON

Analysis: Despite adding structural redundancy via multiple spanning trees, this configuration increased the convergence time while keeping the tracking error essentially unchanged. This suggests that overly complex or dense networks do not necessarily guarantee superior performance within a state-feedback consensus framework.

Different Trajectory - Prestational Analysis

Experiment 11 - Circular Trajectory:

- **Topology:** Undirected ($Adj = [0, 1, 1; 1, 0, 1; 1, 1, 0]$)
- **Trajectory:** Circular
- **Saturation & Noise:** ON

Analysis: Introducing a circular trajectory poses a higher dynamic complexity due to the continuous variation in the velocity vector. This resulted in a slight degradation in overall performance, driving Agents 1 and 2 to frequently hit their physical saturation limits just to manage the continuously changing centripetal requirements.

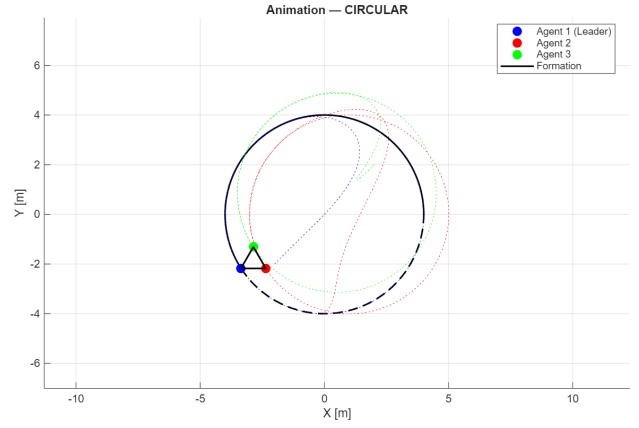


Fig. 6: Performance for a circular trajectory.

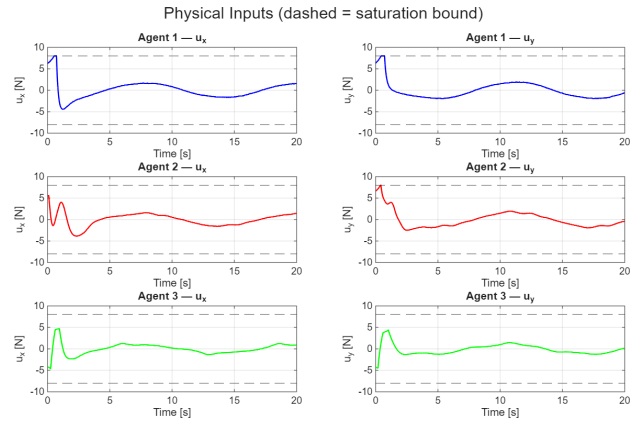


Fig. 7: Physical inputs for a circular trajectory.

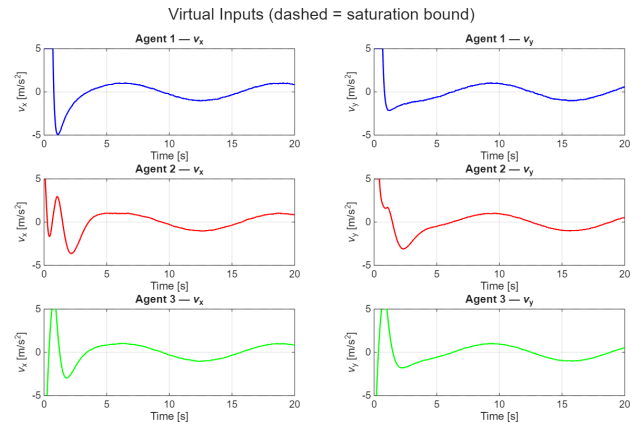


Fig. 8: Virtual inputs for a circular trajectory.

Experiment 12 - Sinusoidal Trajectory:

- **Topology:** Undirected ($Adj = [0, 1, 1; 1, 0, 1; 1, 1, 0]$)
- **Trajectory:** Sinusoidal
- **Saturation & Noise:** ON

Analysis: The sinusoidal path proved to be the most critical operational scenario for the control architecture. The aggressive, time-varying oscillatory nature of the path imposed

continuous dynamic demands, causing heavy saturation of both the virtual and physical control inputs. The combination of aggressive trajectory dynamics and stringent input constraints clearly limits the system's ability to maintain high precision, suggesting that tracking complex curves requires less conservative saturation bounds or elevated coupling gains.

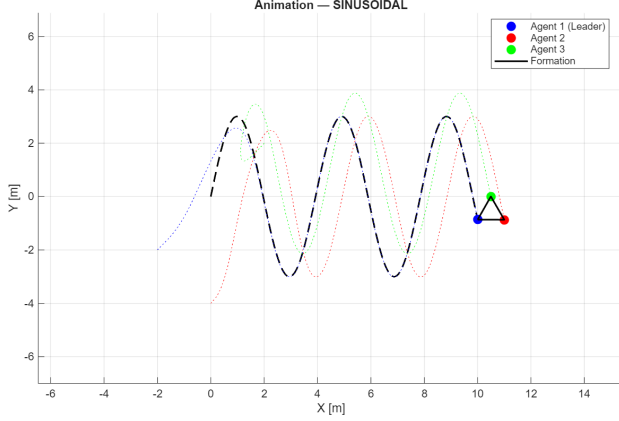


Fig. 9: Performance for a sinusoidal trajectory.

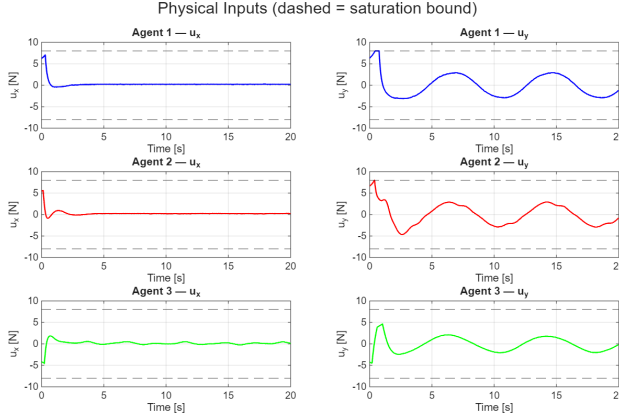


Fig. 10: Physical inputs for a sinusoidal trajectory.

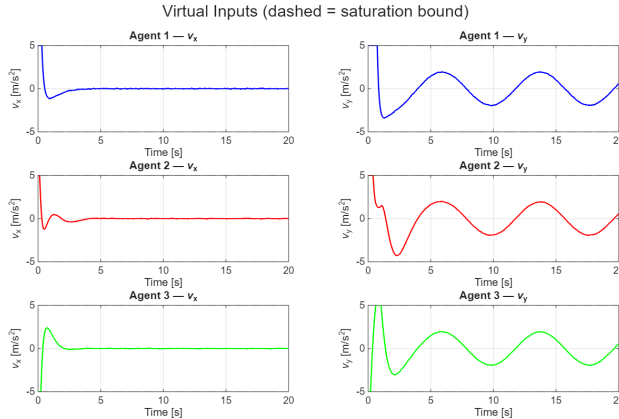


Fig. 11: Virtual inputs for a sinusoidal trajectory.

IV. DISTRIBUTED CONTROL: LOOP-SHAPING APPROACH

The second cooperative control strategy investigated is based on the Loop-Shaping approach. This frequency-domain methodology relies exclusively on the relative position errors measured among neighboring agents. The core concept is to design a dynamic output-feedback compensator $K(s)$ capable of stabilizing the entire multi-agent network by analyzing its modal decomposition.

A. Controller Design

Thanks to the exact Input-Output Feedback Linearization described in Section II, the heterogeneous nonlinear dynamics of the fleet are homogenized into identical linear double integrators, i.e., $G(s) = 1/s^2$.

According to the modal decomposition principle for multi-agent systems, the complex interconnected network can be decoupled into N independent Single-Input Single-Output (SISO) virtual plants. The open-loop transfer function for the i -th mode of the network is given by:

$$L_i(s) = \mu_i G(s) K(s) = \mu_i \frac{1}{s^2} K(s) \quad (1)$$

where μ_i are the eigenvalues of the pinned Laplacian matrix $L_p = L + \gamma \Pi_1$. Since the communication graph contains a spanning tree rooted in the leader, all eigenvalues μ_i have a strictly positive real part. The control objective is to design a single compensator $K(s)$ that simultaneously satisfies the Nyquist stability criterion for all $L_i(s)$ loops.

From PD to Lead Compensator: The presence of a double integrator ($1/s^2$) implies an intrinsic phase of -180° . A purely proportional controller would lead to marginal stability or instability. To lift the phase and guarantee a sufficient Phase Margin (PM), a derivative action is strictly required. A theoretical Proportional-Derivative (PD) controller is defined as:

$$K_{PD}(s) = K_p + K_d s \quad (2)$$

While a PD controller introduces a zero that shifts the phase towards -90° ensuring stability, it is not physically realizable for cyber-physical systems subject to sensor noise. The pure derivative term amplifies high-frequency measurement noise to infinity, leading to instantaneous actuator saturation.

To overcome this structural limitation, a high-frequency pole is added, transforming the PD into a *Lead Compensator* (Filtered PD):

$$K_{Lead}(s) = K_p + \frac{K_d s}{1 + \tau_{lead} s} \quad (3)$$

The zero provides the necessary phase advance around the crossover frequency (ω_c), while the pole at $\omega_p = 1/\tau_{lead}$ attenuates the high-frequency gain, protecting the actuators from noise amplification. Additionally, to improve the tracking of time-varying trajectories (e.g., sinusoidal paths) without unacceptable phase lag, a feedforward action based on the reference acceleration $\ddot{r}(t)$ is added to the control law.

B. Implementation and Workflow

The continuous-time simulation of the Loop-Shaping protocol is implemented in MATLAB using the `ode45` solver. Since the Lead Compensator introduces a dynamic element (the low-pass filter), the state space of the system must be augmented.

The time-domain equivalent of the high-frequency pole is implemented by adding two new states per agent ($z_{x,i}, z_{y,i}$), increasing the total state vector from 12 to 18 variables. The filter dynamics are computed at each integration step as:

$$\dot{z}_i = \frac{K_d \dot{\epsilon}_i^p - z_i}{\tau_{lead}} \quad (4)$$

where $\dot{\epsilon}_i^p$ is the distributed relative velocity error computed via the Laplacian matrix. Consequently, the virtual control input ν_i applied to the IOFL block is evaluated as:

$$\nu_i = K_p \epsilon_i^p + z_i + \ddot{r}(t) \quad (5)$$

This architecture allows a seamless switch between the ideal PD controller (by setting $\dot{z}_i = 0$ and using raw derivative error) and the robust Lead Compensator, enabling the comparative experimental analysis presented in the next subsection.

C. Method-Specific Experiments and Analysis

To validate the controller, a critical analysis of the compensator synthesis was conducted. The tuning process was driven by the trade-off between trajectory tracking, disturbance rejection, and measurement noise attenuation.

1) The Ideal Scenario and the Failure of the PD Controller:

In a purely theoretical scenario without external disturbances, a standard PD controller successfully drives the fleet into the desired triangular formation. To establish a baseline, this standard PD architecture was initially tested without feedforward and without measurement noise. Under these ideal conditions, the graphical results confirmed a perfect transient response and zero steady-state error, proving the theoretical validity of the control law.

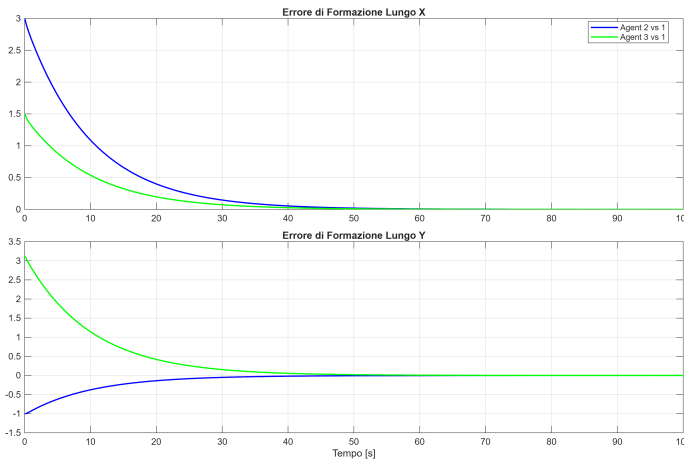


Fig. 12: Formation Error in the ideal case (without noise and disturbances) using a pure PD controller

However, real-world cyber-physical systems rely on sensors affected by high-frequency noise. By activating a simulated sensor noise at $\omega \approx 100$ rad/s (`p.USE_NOISE = true`), the limitations of the un-filtered PD controller became immediately apparent. The pure derivative action amplified the noise, causing the physical control inputs (u_x, u_y) to violently and continuously hit the saturation bounds (± 8 N). This chattering effect destroys the formation stability and would rapidly lead to mechanical failure of the real actuators.

2) Frequency Domain Synthesis of the Lead Compensator:

To solve the noise amplification issue without losing the stability guaranteed by the PD zero, the Lead Compensator was synthesized using Bode plots. The analysis focused on two critical transfer functions:

- **Sensitivity Function** $S(s) = 1/(1 + L_i(s))$: Defines the ability to track references and reject low-frequency disturbances (e.g., constant wind). Since the plant is a Type-2 system (double integrator), $S(s) \rightarrow 0$ for $\omega \rightarrow 0$, ensuring perfect steady-state tracking of linear ramps.
- **Complementary Sensitivity** $W(s) = L_i(s)/(1 + L_i(s))$: Defines the susceptibility to high-frequency sensor noise.

By choosing a filter time constant $\tau_{lead} = 0.02$ s, a pole was strategically placed at $\omega_p = 50$ rad/s. As illustrated in the Bode plots in Fig 14, this pole forces the magnitude of $W(s)$ to drop significantly below 0 dB for frequencies higher than 50 rad/s. Specifically, at the noise frequency of 100 rad/s, the attenuation is sufficient to prevent actuator saturation. Simultaneously, the gains were tuned to $K_p = 1$ and $K_d = 10$ to maintain a crossover frequency ω_c that guarantees a Phase Margin of at least 55° for all eigenvalues μ_i , avoiding the -180° instability region.

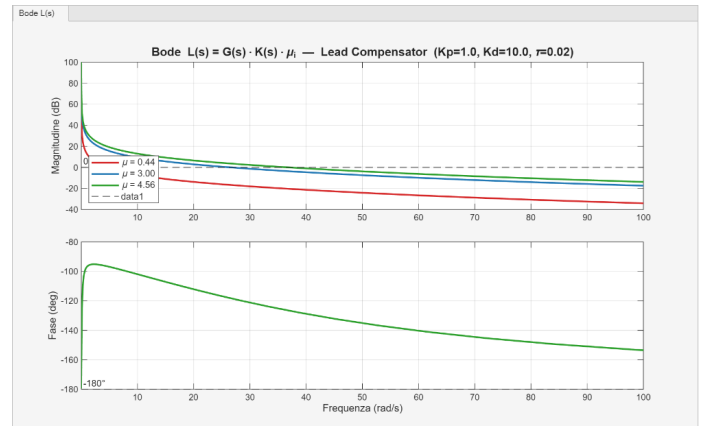


Fig. 13: Bode phase graph of $L(S)$ showing the marginal phase in ω_c

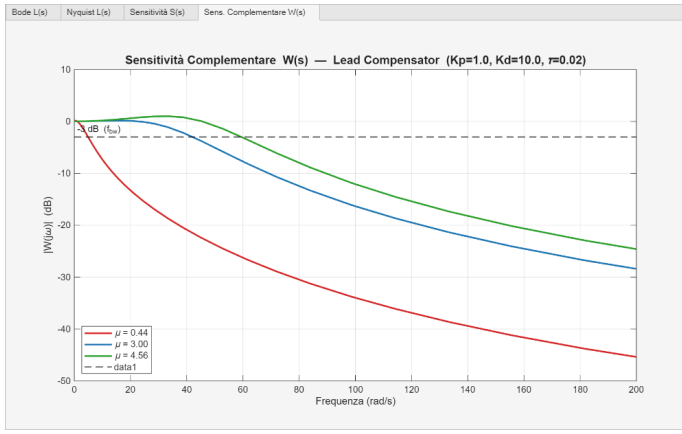


Fig. 14: Complementary Sensitivity ($W(s)$) of the final controller (without feedforward)

At this stage, the Lead Compensator was implemented strictly without any feedforward action. The final controller constructed shows satisfying performances in the linear case trajectory configuration.

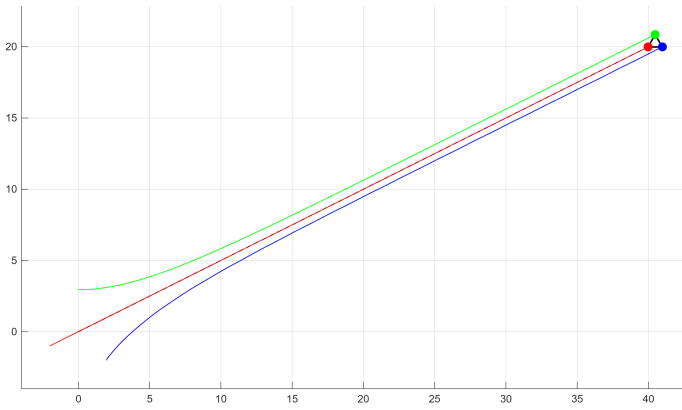
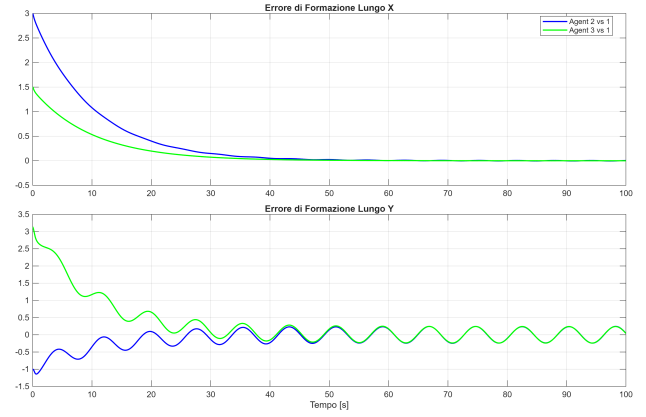
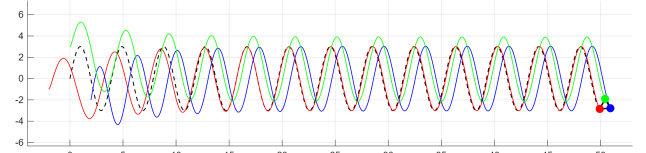


Fig. 15: Final controller acting on agents following a linear trajectory.

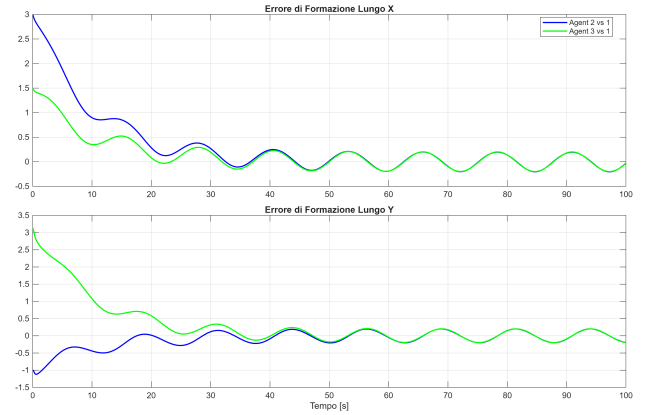
While the added pole at 50 rad/s successfully manages the high-frequency noise and protects the actuators, relying solely on feedback introduces a new limitation. When testing this feedback-only configuration against non-linear references, such as circular or sinusoidal trajectories, the formation exhibits a noticeable tracking lag. Because the compensator reacts only after a position error has accumulated, the system fails to perfectly synchronize with accelerating trajectories, as demonstrated by the tracking delay below.



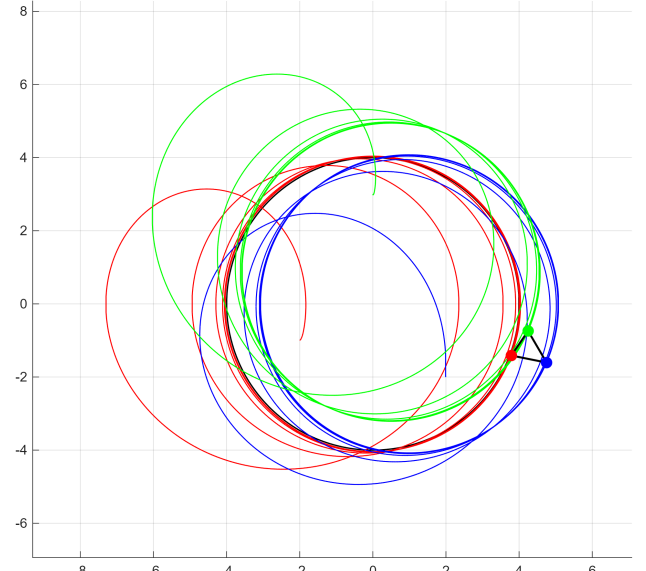
(a) Formation error – sinusoidal trajectory



(b) Final trajectory – sinusoidal



(c) Formation error – circular trajectory



(d) Final trajectory – circular

Fig. 16: Formation error and agent trajectories for the sinusoidal (top) and circular (bottom) reference cases.

3) *Feedforward implementation - definitive solution*: To resolve the non-linear tracking limitations without compromising the noise-rejection properties of the Lead Compensator, a feedforward term was added. By injecting the nominal trajectory acceleration ($\ddot{r}$) directly into the control action, the feedforward anticipates the dynamic effort required for non-linear maneuvers.

The synthesized Lead Compensator was tested in the time domain under aggressive conditions: simultaneous sinusoidal trajectory tracking, high-frequency sensor noise (100 rad/s), and a low-frequency disturbance (5 rad/s) applied to Agent 2.

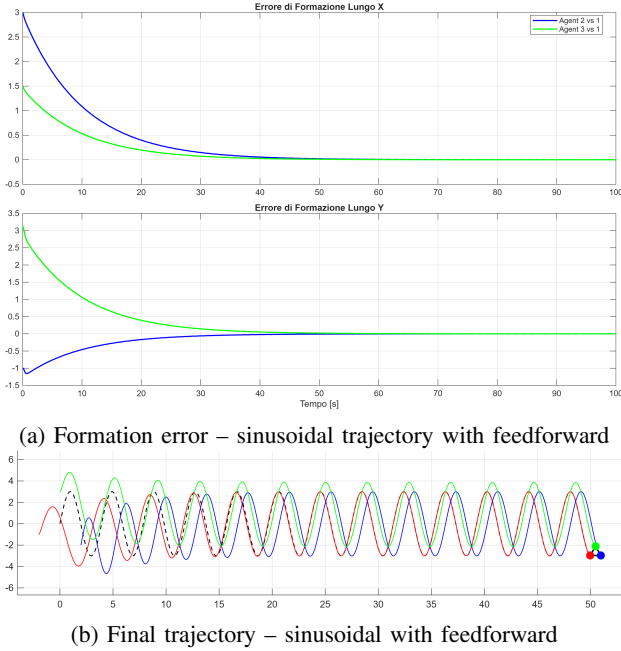


Fig. 17: Formation error and agent trajectory for the sinusoidal case with added feedforward term.

The results fully validate the frequency-domain design. The disturbance at 5 rad/s is effectively attenuated because the magnitude of the Sensitivity function $S(s)$ at this specific frequency is sufficiently low, keeping the residual formation error tightly bounded. More importantly, the high-frequency sensor noise is successfully filtered by the compensator pole (via $W(s)$), resulting in smooth, continuous physical control inputs u_i well within the ± 8 N saturation limits.

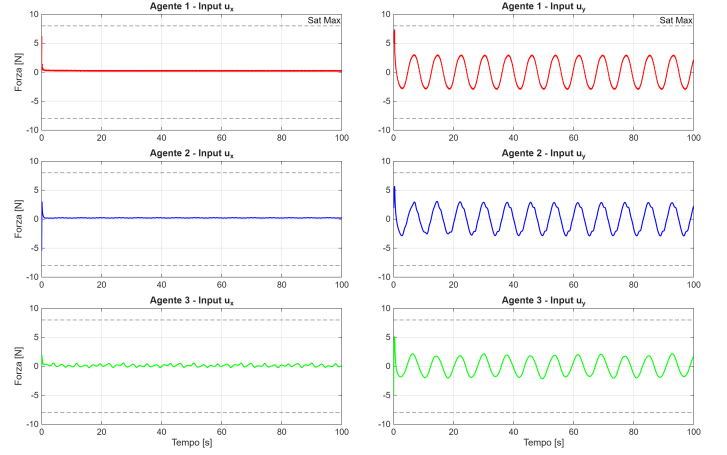


Fig. 18: Plot of the saturation limits showing the corrected behaviour of the bounded input not exceeding the imposed thresholds

By combining the lead compensator with the feedforward action, the final control architecture significantly reduces the tracking lag observed for the sinusoidal and circular trajectories, ensuring accurate trajectory tracking while maintaining robustness against sensor noise and avoiding input saturation.

V. COMPARATIVE EXPERIMENTAL ANALYSIS

To provide an evaluation of the proposed architectures, this section directly compares the State Variables Feedback (SVFB) and the Loop-Shaping approaches.

A. Informational Requirements and Architecture

The SVFB protocol is inherently data-intensive. Computing the control law requires the full state vector. Since sensors provide only position data, this necessitates a Luenberger Observer for velocity estimation, increasing the architectural complexity of the individual agent.

Conversely, the Loop-Shaping approach is computationally leaner. As a pure output-feedback protocol, it computes the control action relying exclusively on relative position errors. The necessity to estimate velocities is completely bypassed by the internal dynamics of the Lead Compensator.

B. Tuning Methodology and Stability Guarantees

SVFB tuning relies on the LQR energetic balance defined by the Q and R matrices so it requires an iterative, trial-and-error process.

The Loop-Shaping approach offers a strictly analytical frequency-domain synthesis. Controller parameters (K_p , K_d , τ_{lead}) are deterministically calculated via Bode and Nyquist diagrams. In the proposed design, a Phase Margin of at least 55° was mathematically guaranteed *a priori* for all network eigenvalues, ensuring robust stability.

C. Disturbance and Noise Rejection Capabilities

The divergence in performance is most evident in the presence of external anomalies. In Experiment 8, the SVFB

controller exhibited severe degradation against a sinusoidal output disturbance.

Loop-Shaping inherently manages these phenomena through the active shaping of its sensitivity functions, keeping the residual formation error caused by the disturbance bounded to negligible values, outperforming the SVFB in noisy environments.

D. Trajectory Tracking and Actuator Effort

Both controllers guarantee zero steady-state tracking error for linear trajectories, but exhibit a structural tracking lag for accelerating references (circular and sinusoidal paths).

Both architectures require a kinematic feedforward term ($\ddot{r}$) to resolve this lag. Once augmented, the SVFB method achieves slightly faster transient responses, but this rapidity causes frequent actuator saturation, reaching the physical limit of 8 N during sharp curves. The Loop-Shaping Lead Compensator imposes a smoother dynamic response, successfully preventing actuator chattering while keeping the tracking error bounded

VI. CONCLUSION

This project successfully addressed the problem of distributed formation control for a heterogeneous network of planar mobile robots.

The State Variables Feedback (SVFB) protocol demonstrated excellent trajectory tracking capabilities and rapid convergence in ideal conditions. Its reliance on full-state estimation and its high sensitivity to disturbances limit its applicability in harsh environments.

The Loop-Shaping output-feedback approach proved to be a highly reliable solution. The frequency-domain synthesis of a Lead Compensator, coupled with kinematic feedforward, protected the actuators and provided robust stability and noise attenuation.

Ultimately, the SVFB method is suitable for highly predictable environments with stringent tracking speed requirements and precise sensors, whereas the Loop-Shaping approach is the strictly superior and robust alternative for cyber physical systems operating in noisy and disturbed real-world scenarios.